

TLS_Pie Raspberry Pi Setup Checklist

Use this checklist to set up the Raspberry Pi for the TLS_Pie lidar capture system.

1. Prepare the Pi

Insert the microSD card with Raspberry Pi OS.
Boot the Pi and log in.
Connect the Pi to the network.
Open a terminal.

2. Run the setup script

Run this command:

```
sudo bash /home/lipi/TLS-Pie/Raspberry\Pie4/setup_tls_pie_pi.sh
```

If the project is not yet in /home/lipi/TLS-Pie, copy it there first.

3. Verify the required packages are installed

The script installs:

python3-pip
python3-dev
python3-rpi.gpio
git
tcpdump
openssh-server

4. Verify the project files are present

Make sure these files exist:

/home/lipi/TLS-Pie/Raspberry\Pie4/TLS-Pie/VLPrecord.sh
/home/lipi/TLS-Pie/Raspberry\Pie4/TLS-Pie/VLPbuttons.py
/home/lipi/TLS-Pie/Raspberry\Pie4/TLS-Pie/VLPwaitbutton.py

5. Test the script manually

Run:

```
sudo /home/lipi/TLS-Pie/Raspberry\Pie4/TLS-Pie/VLPrecord.sh eth0
```

Expected behavior:

the script starts
it waits for the Arduino trigger
it creates a .pcap file in /home/lipi/velodyne

6. If the interface is not eth0

Check the interface name:

```
ip -br addr
```

Then run the script with the correct interface name.

7. Enable automatic startup

The setup script creates a startup launcher for the Pi desktop session.

Reboot the Pi:

```
sudo reboot
```

8. Confirm the system works

After reboot:

verify the script starts automatically

verify a .pcap file is created

verify the Pi is waiting for the Arduino trigger

9. Common problems

Wrong network interface name

Missing packages

No common ground between Arduino and Pi

No .pcap file created

10. Quick recovery commands

If needed, run:

```
sudo apt update
```

```
sudo apt install -y python3-pip python3-dev python3-rpi.gpio git tcpdump openssh-server
```

```
sudo pip3 install RPi.GPIO
```